Chia-Heng Liu

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Education

Cornell University Ithaca, NY
Master of Engineering in Electrical and Computer Engineering *Exp. Dec. 2026*
Topic: Mixed-Precision LLM Inference on Next-Generation AI Accelerators Advisor: Prof. Zhiru Zhang
Coursework: Computer Architecture, Data Center Architecture, Computer Networks and Telecommunications

National Tsing Hua University (NTHU) Hsinchu, Taiwan
Bachelor of Science in Materials Science and Engineering *Sep. 2020 – Jun. 2024*
Coursework: Logic Design, Computer Organization, Electronics, Signals and Systems, Semiconductor Design

Experience

Research Assistant at NYCU CERES Lab *Jul. 2024 – Aug. 2025*
Advisor: Prof. Kun-Chih (Jimmy) Chen
Publication (First Author): Energy-Efficient and Accurate Stochastic Computing-based Multiply-Accumulate Architecture for Neural Network Accelerator Design *Feb. 2025 – Jul. 2025*
To appear at IEEE International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoC), Dec. 2025

- Designed a novel SC-based Multiply-Accumulate architecture for energy-efficient deep neural network (DNN) inference on edge devices.
- Achieved up to 90.6% reduction in mean squared error (MSE) over prior SC-based MAC designs, with synthesis in TSMC 40 nm technology demonstrating compact area (1049 μm^2) and low power (0.645 mW).
- Maintained 96.76% classification accuracy on MNIST when deployed in a SC neural network, matching binary performance (97.06%) with significantly reduced resource cost.

Co-op Project with ITRI: NoC Accelerator Simulator Design and Low-Power Data Transmission Protocol, Movement, and Reuse Method *Jul. 2024 – Jan. 2025*

- Enabled NoC latency instrumentation and RISC-V workload execution by modifying the OpenPiton multicore simulator to support both C and assembly inputs, enhancing observability of data transfers.
- Built a full-system AI accelerator simulator integrating Gem5 (RISC-V CPU) and Noxim (NoC) with MCU-based task scheduling across PEs and cache-interconnect modeling.
- Enhanced NoC simulation with multi-layer architectures and data dependency-aware transmission protocols; evaluated system performance with neural network workloads to identify bottlenecks and optimize data reuse.

Selected Projects

ASIC Project: Systolic CNN Accelerator with Stepwise Weight Streaming for Maximized PE Utilization
System Verilog, Synopsys (VCS, Verdi, Design Compiler), ARM (SRAM Compiler), Pytorch *May. 2025 – Jul. 2025*

- Applied tiling and scheduling with fused Max–ReLU–MAC operations, achieving **129 GFLOPs/s (INT8-equivalent) at 500 MHz**, and optimized parallel activation processing to improve efficiency across convolutional neural networks.
- Designed and implemented a CNN accelerator leveraging a systolic array with a cached weight dataflow, enabling cyclic weight movement within PEs to maximize reuse and minimize memory access.
- Implemented on-chip SRAM buffering to reduce off-chip memory access latency and lower bandwidth pressure within the processor–memory dataflow.
- Completed ASIC flow from logic synthesis to gate-level verification using Synopsys tools (Design Compiler targeting TSMC 40 nm, VCS, Verdi), achieving a synthesized cell area of **412.7K μm^2**.
- Verified correctness with RTL simulations and deployed a quantized MNIST model on a custom-ISA accelerator, validating full-stack functionality.

Technical Skills

Hardware: Verilog, SystemVerilog, SystemC, RISC-V, Network-on-Chip, OpenPiton
EDA Tools: Synopsys (VCS, Verdi, Design Compiler, DesignWare), ARM (SRAM Compiler), Cadence (Innovus), Intel ModelSim, Xilinx Vivado, Verilator
Programming: C++, Python, Pytorch, RISC-V Assembly, Linux, Shell, TCL, LaTeX